



ITU Electrical-Electronics Faculty – Electrical Engineering Department

ELECTRICAL DISTRIBUTION AUTOMATION LABORATORY

LAB REPORT

Experiment No and Title	6: Photovoltaic Solar Cells
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Experiment date	05.03.2025
Submission date of the lab report	12.03.2025
Grade	

The Purpose of the Experiment

The experiments are designed to explore the connection between the solar energy received by the panels and the electrical energy produced at the terminals of the solar panels, in addition to examining the parameters that affect this conversion.[1]

General Information Related to the Experiment

The interaction of light with a photovoltaic cell, often termed a solar cell, can result in one of three outcomes: the light may be reflected, absorbed, or it may traverse the cell without being impeded [2]. The structure of a photovoltaic cell includes semiconductor materials which are capable of transforming sunlight directly into electrical power. Numerous semiconductor materials are utilized in the construction of pv cells. When light interacts with a semiconductor, the material absorbs the energy from the light and transmits it to the negatively charged particles. The presence of this extra energy permits the electrons to navigate through the material, leading to the establishment of an electrical current [2].

Solar cells are designed to harness the pv effect, wherein photons of light interact with electrons, energizing them to a higher state. This increase in energy permits the electrons to act as charge carriers, thus enabling the production of electric current. The efficiency of converting solar energy into electrical energy varies between %5 and %20. A crystalline silicon solar cell produces a direct current output voltage of 0.58 volts.

The I-V characteristic curves associated with solar cells represent the current and voltage dynamics of PV cell. It elucidates its performance in solar energy conversion, highlighting both its capabilities and efficiency metrics. It is crucial to ascertain the I-V characteristics, with particular emphasis on the P_{max} , of a solar cell in order to accurately evaluate its performance and solar efficiency. The information derived from I-V curves is important for the proper configuration of a solar system, allowing it to operate in proximity to its optimal peak power point. Consequently, the most effective performance of a photovoltaic cell, panel, is established when it operates at the maximum power point [3].

The relationship between temperature and photovoltaic cell performance is a consequence of the semi-conductive characteristics of these cells. Efficiency remains unaffected by low temperatures, whereas high temperatures lead to a significant reduction in productivity. At high temperatures, the efficiency of pv cells fabricated from monocrystalline silicon surpasses that of those made from polycrystalline silicon. Output power of pv cells diminishes as the temperature of the cell rises [4]

Operating on the principles of the thermoelectric effect, a Peltier element enables the direct conversion of temperature variations into electrical voltage and can also reverse this process. In a system that operates on the Peltier effect, the electrodes are attached to a sample composed of semiconductor material. Upon connecting a DC voltage source to the electrodes, the side that carries a negative charge is observed to cool down, in contrast to the side that carries a positive charge, which tends to heat up. The arrangement involves placing the negative electrode in contact with the component, device or medium intended for cooling, while the positive electrode is connected to a heat sink that facilitates the radiation or dissipation of thermal energy into the ambient environment.

The Tools Used in the Experiment

Monocrystalline solar cell

Polycrystalline solar cell

HID lamp - small solar panel

Peltier element

Pyranometer

Amperemeter

Voltmeter

Step-up and Step-down Converter

Battery to retain excess energy generated by solar cells

Fuse

Load to draw energy from the output of solar cells.

Monitor to read panel temperature

USB Thermometer

Temperature sensor

Fan for cooling down

Microcontroller unit to manage temperature controlling.

Wattmeter

The Interpretation of the Experiment Results

Elevated irradiance levels cause the curve to shift upward, resulting in increased power generation from the photovoltaic module. As depicted in the figure, there exists a straightforward association between the augmentation of radiation intensity and the increase in short-circuit current. The relationship between voltage and radiation intensity becomes logarithmic when the strength of the radiation is amplified.

It is seen from the graph "Temperature vs Solar Irradiation" that ideal panel power decreases with increasing temperature and is higher when solar irradiation increases for the same temperature value. The rise in temperature contributes to a diminished ideal power generation.

PROCESSES DURING THE EXPERIMENT

1)

Solar radiation intensity = 396 [W/m²]

Panel temperature = 40 [°C]

Tip	P _{max} [W]	V _{max} [V]	I _{max} [A]
Monocrystalline	21	13	1.6
Polycrystalline	43.25	33	1.3
Thin Film	—	—	—

Total P_{max} = 64.25 [W]

P_{max}(tot) = 41 [W]

Converter output voltage =	27.1	[V]
Converter output current =	2.07	[A]
Converter output power =	56.1	[W]
Load current =	0.96	[A]
Battery voltage =	26.5	[V]
Battery charge current =	2.04	[A]
Battery discharge current =	0	[A]

2) 25°C

L=1000 [W/m ²]			L=750 [W/m ²]		
Voltage [V]	Current [mA]	Power [W]	Voltage [V]	Current [mA]	Power [W]
0	I _{sc} : 18.4	0	0	I _{sc} : 16.9	0
2.02...	39.4	80	2.02...	13.7	27
4.04...	36.9	149	4.04...	11.4	43
6.06...	39.6	239	6.06...	28.3	170
8.08...	36.9	296	8.08...	25.	202
10.10...	32.2	324	10.10...	21.7	220
12.12...	26.6	321	12.12...	18.6	224
V _m : 10.83	I _m : 29.19	P _{max} : 326	V _m : 12.07	I _m : 18.6	P _{max} : 225
14.14...	21.30	301	14.14...	14.1	199
16.16...	14.30	230	16.16...	9.7	155
18.18...	3.5	136	18.18...	4.8	87
V _{oc} : 20.03	0	0	V _{oc} : 19.77	0	0

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L=500 [W/m ²]			L=250 [W/m ²]		
Voltage [V]	Current [mA]	Power [mW]	Voltage [V]	Current [mA]	Power [mW]
0	I _{sc} : 24	0	0	I _{sc} : 12.7	0
2.02...	23.30	48	2.02...	12.8	26
4.04...	22.30	89	4.00...	12.5	50
6.01...	20.40	125	6.01...	12.2	73
8.03...	18.40	148	8.03...	12	96
10.05...	16.30	164	10.00...	11.4	116
12.07...	13.60	164	12.00...	10.1	120
V _m : 12.23	I _m : 13.90	P _{max} : 171	V _m : 12.8	I _m : 9.5	P _{max} : 122
14.03...	11.00	154	14.03...	8.2	114
16.05...	7.10	115	16.00...	6	96
18.01...	3.10	55	18.01...	1.7	31
V _{oc} : 19.42	0	0	V _{oc} : 18.74	0	0

L=1000 [W/m²] ; I_{sc} = 410.8 [mA] for T = 40 [°C] L=500 [W/m²] ; I_{sc} = 241.2 [mA] for T = 40 [°C]

Temperature [°C]	V _{oc} [V]	P _p [mW]	Temperature [°C]	V _{oc} [V]	P _p [mW]
25	20.09	819.672	55	16.78	406.076
30	19.81	808.248	50	17.28	418.176
35	19.30	787.144	45	17.96	434.632
40	18.86	769.488	40	18.46	446.732
45	18.46	753.168	35	18.91	457.622
50	17.96	732.768	30	19.36	468.512
55	17.51	714.408	25	19.59	474.078

- Open circuit voltage (V_{oc}) : 20.03 [V]
- Short circuit current (I_{sc}) : 18.4 [A]
- Ideal panel power (P_p) : 819.67 [mW]
- Voltage at maximum power (V_{mp}) : 10.83 [V]
- Current at maximum power (I_{mp}) : 29.9 [A]
- Panel power (P_{max}) : 326 [mW]
- Panel efficiency (η_p) : [%]
- Temperature power factor (α_p) : [%]

b)

Formulas to be used when calculating cell and panel efficiency:

$$\eta = \frac{V_m I_m}{P_{in}} = \frac{P_{max}}{P_{in}}, \quad P_{in} = L_{sun} \cdot S_{cell}, \quad L_{sun} = 976 \text{ W/m}^2$$

For cell efficiency $\eta_c \rightarrow P_{max} = P_{stc}$

$$P_{stc} = \frac{P^T}{1 + (T - 25) \alpha_p} \quad T = 40^\circ\text{C} \quad P_{in} = L_{sun} \cdot \underbrace{S_{cell}}_{\substack{\text{includes} \\ \text{fill factor} \\ \text{for } \eta_c}}$$

For panel efficiency η_p :

P_{max} is the measured data

For mono

$$\eta_{c,mono} = \frac{P_{stc}}{P_{in}} \quad P_{stc} = \frac{21}{1 + (40 - 25) \cdot \frac{-0.5}{100}} = 22.70 \text{ W} \quad P_{in} = 976 \cdot 0.6 \cdot 0.63 \cdot \left[\frac{65}{100} \right] = 239.80 \text{ W}$$

$$\eta_{c,mono} = \frac{22.70}{239.80} = \% 9.46$$

$$\eta_{p,mono} = \frac{21}{P_{in}}$$

$$P_{in}' = 976 \cdot 0.6 \cdot 0.63 = 368.92 \text{ W} \quad \eta_{p,mono} = \frac{21}{368.92} = \% 5.69$$

For poli

$$P_{stc} = \frac{43.25}{1 + (40 - 25) \cdot \frac{-0.47}{100}} = 46.53 \text{ W} \quad P_{in} = 976 \cdot 0.78 \cdot 0.995 \cdot \left[\frac{0.72}{100} \right] = 545.38 \text{ W}$$

$$P_{in}' = 976 \cdot 0.78 \cdot 0.995 = 757.47 \text{ W}$$

$$\eta_{c,poli} = \frac{46.53}{545.38} = \% 8.53$$

$$\eta_{p,poli} = \frac{43.25}{757.47} = \% 5.71$$

c)

In the first step, we found that maximum power with two solar panels is 64.25 W. After connecting two panels in parallel, measured P_{max} dropped to 41 W. This is mainly due to mismatch losses, which occur due to differences in voltage and current characteristics. This leads to a condition where the stronger panel pushes current through the weaker one, creating circulating currents that waste power as heat.

$$e-d) \quad \eta_{converter} = \frac{\text{converter output power}}{\text{measured total } P_{max}} = \frac{56.1}{64.25} = \% 87.31$$

Battery voltage is 26.5 V, so the battery is not fully charged yet.
Battery charge current is 2.04 A indicates active charging. The converter is providing power to both load and the battery.

$$c) P_{in} = I_{sun} \cdot S_{cell} = 1000 \cdot 0,115 \cdot 0,085 = 9,775 W$$

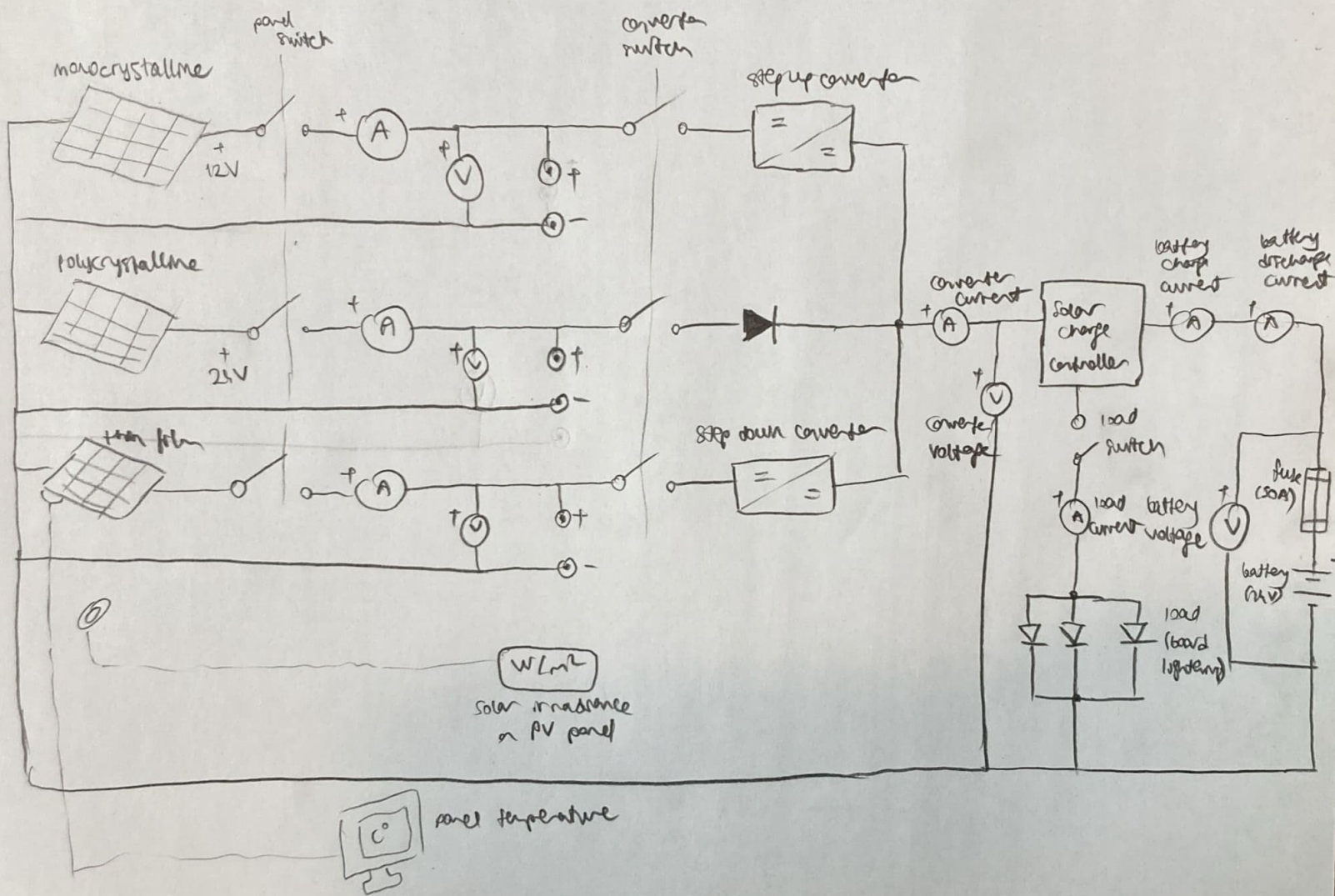
$$P_{max} = 326 \text{ mW (measured)} \quad \eta = \frac{0,326}{9,775} = \% 3,33$$

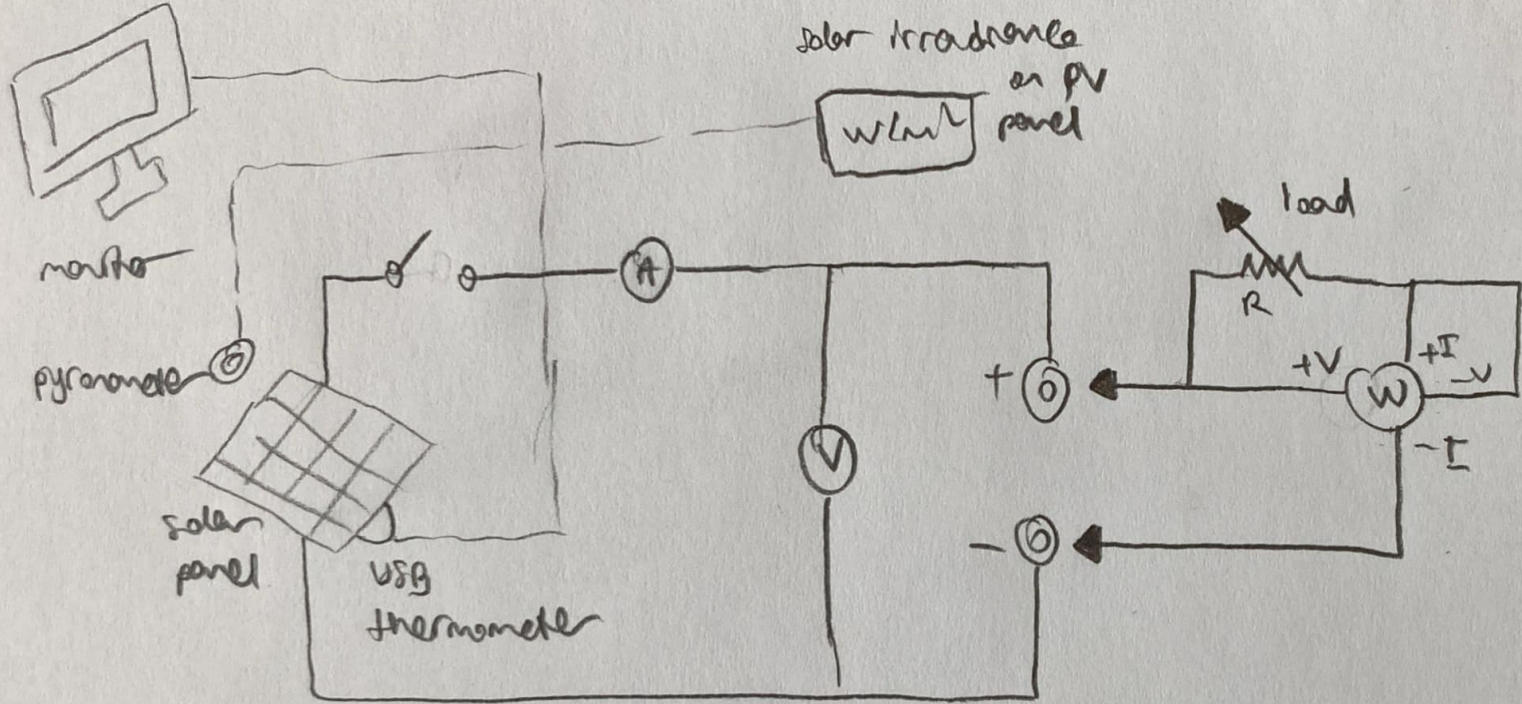
$$P_p = P_{stc} \text{ (} 1000 \text{ W/m}^2 \text{ and } 25^\circ\text{C)} = 819,67 [\text{mW}]$$

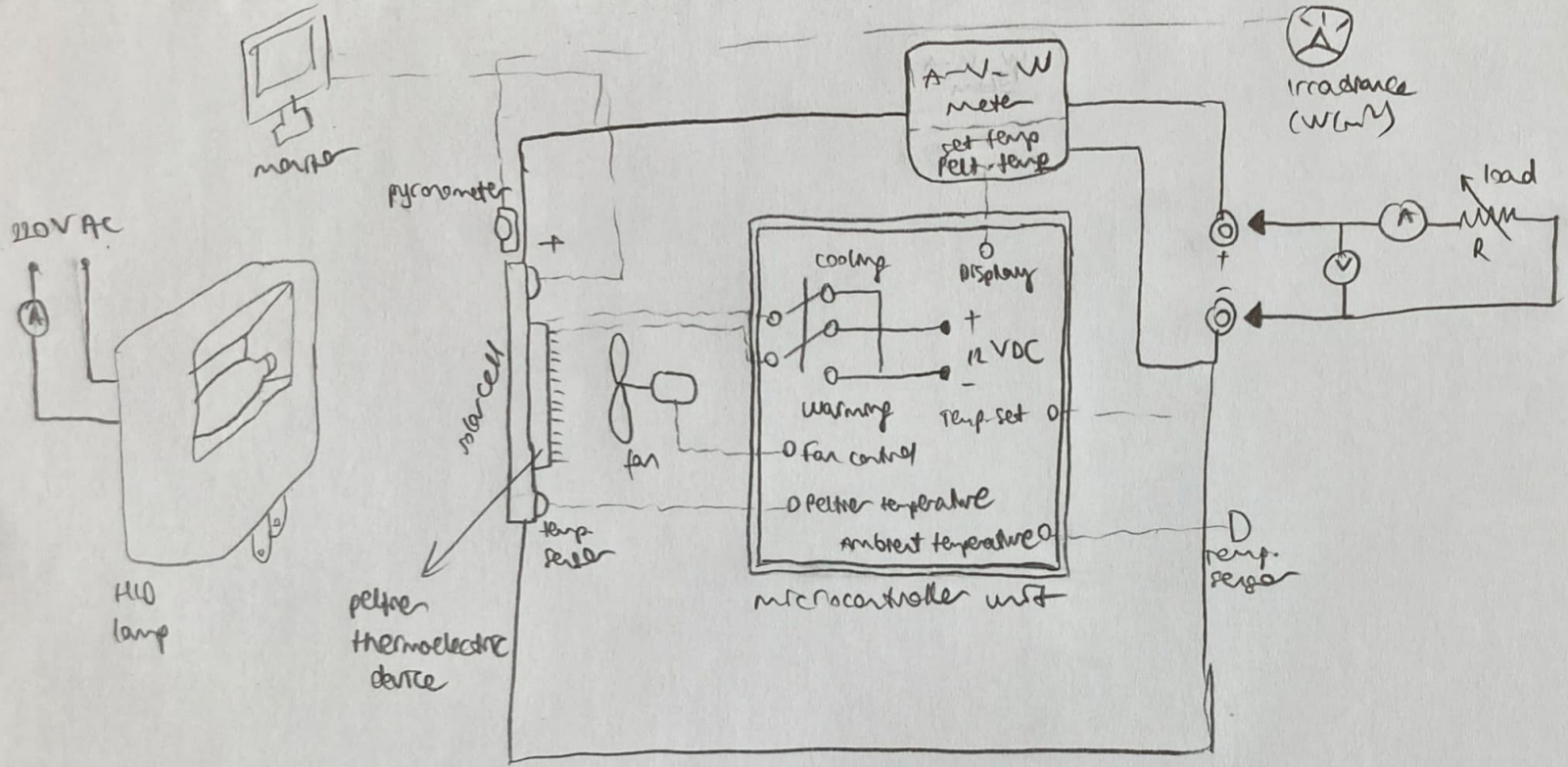
in order to find α_p , let's use the value for 40°

$$P^T = P_{stc} [1 + (T - 25) \alpha_p] \quad 769,488 = 819,67 [1 + 15 \alpha_p]$$

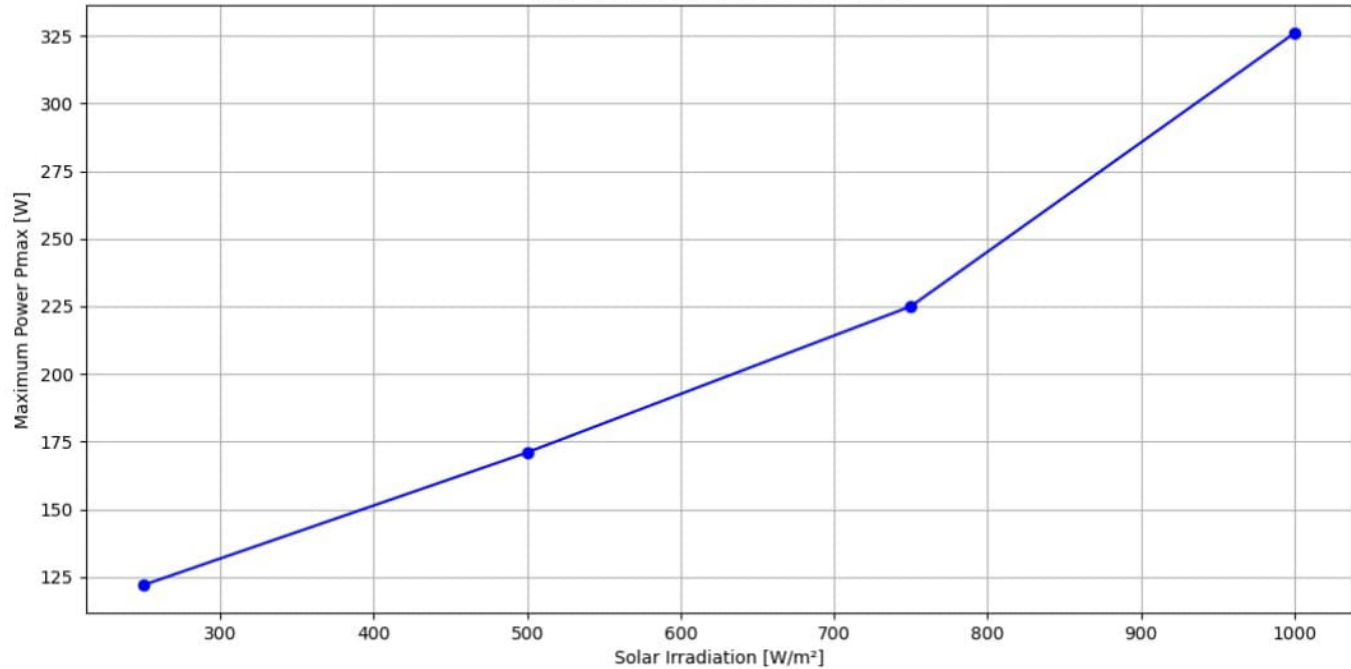
$$\alpha_p = -\% 0,40$$



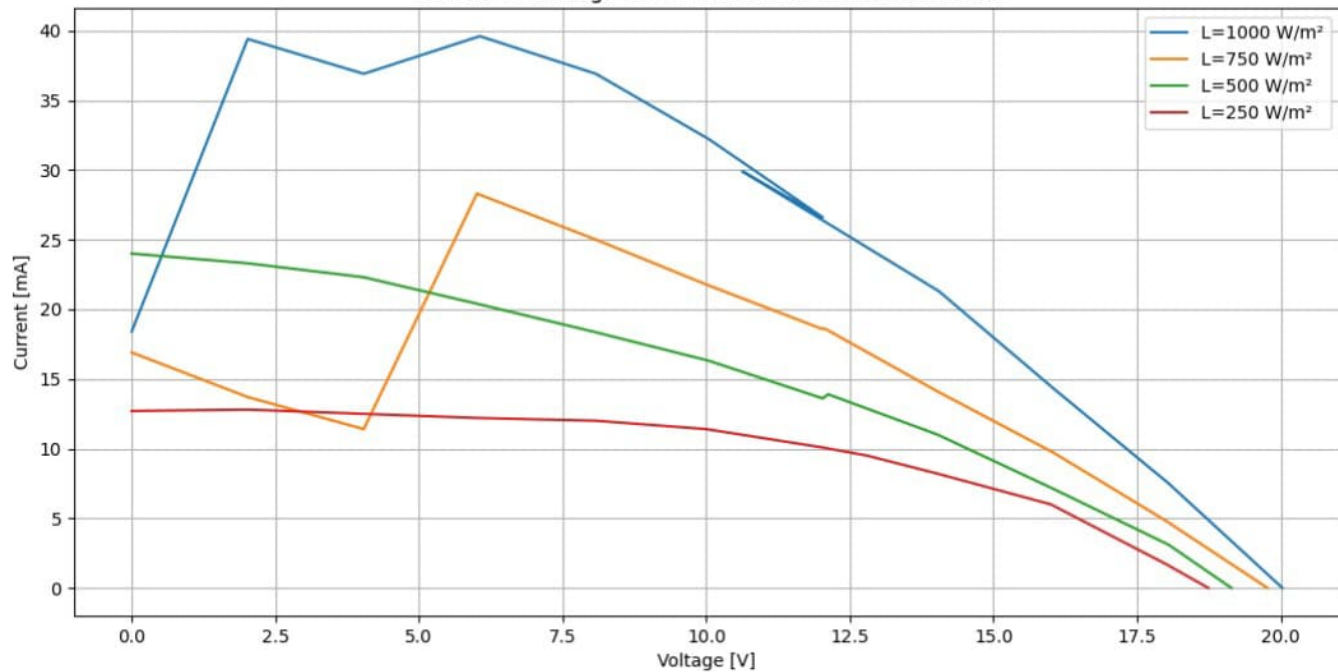




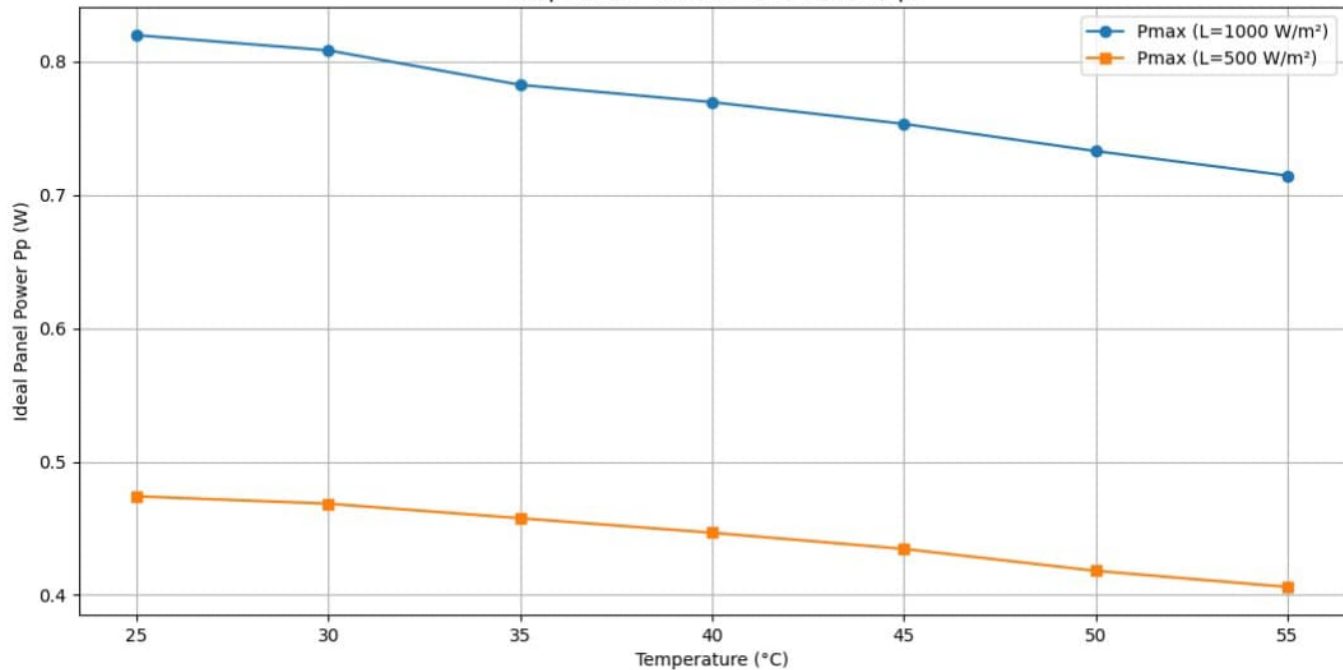
Solar Irradiation vs. Maximum Power



Current vs. Voltage for Different Solar Irradiance Levels



Temperature vs. Ideal Panel Power (P_p)



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